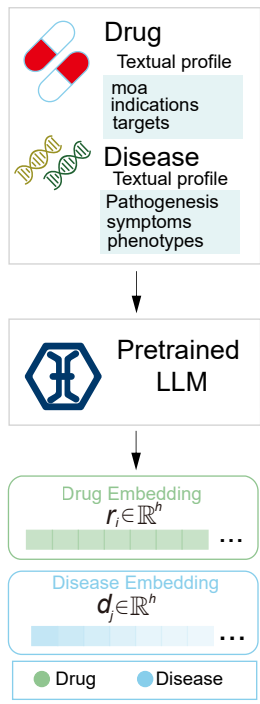
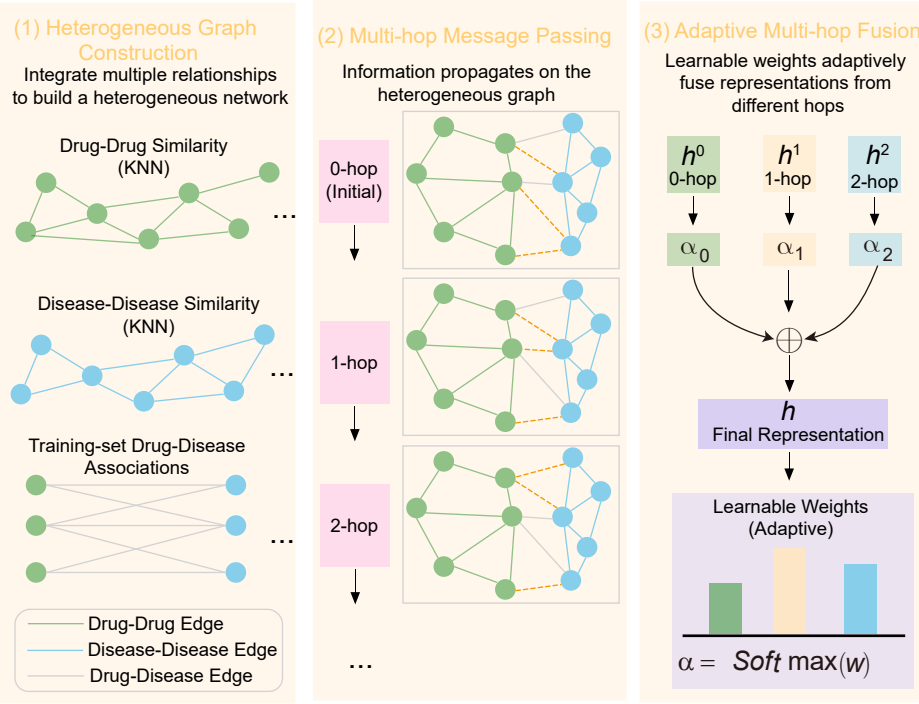


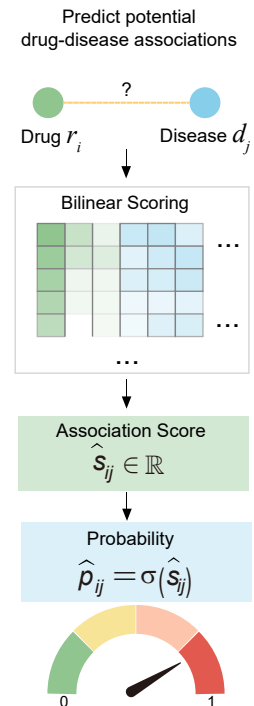
A Semantic Representation LLM



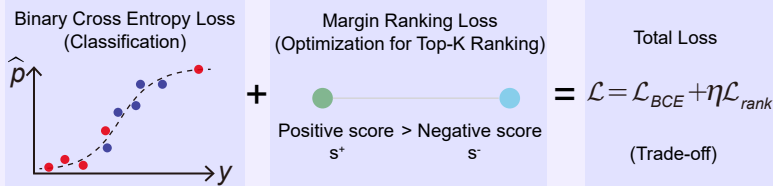
B BiLLM-DR (Bilinear Large Language Model-enhanced Drug Repositioning)



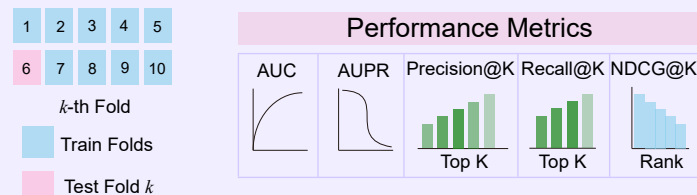
C Prediction



D Training Objective(BCE + BPR)



E Evaluation Protocol (10-fold Cross-Validation)



LLM: Large Language Model KNN: k -Nearest Neighbors BCE: Binary Cross Entropy σ : Sigmoid Function

Green: Drug - related Blue: Disease - related Orange: Drug - Disease Interaction